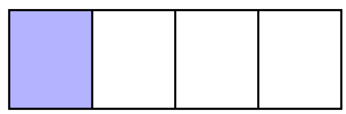
Unit Fractions in Area Models

Directions:

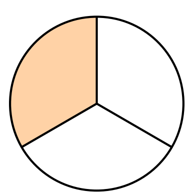
- A **unit fraction** names *one* of the equal pieces a whole is cut into.
- For each model, count how many *equal* pieces make the whole shape. That count is the bottom number of the fraction; the top number stays 1.
- Write fractions in the form $\frac{1}{\square}$, and write your final answer on the answer line.

1. The rectangle below is cut into 4 equal pieces. One piece is shaded. What unit fraction of the rectangle is shaded?



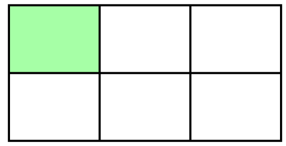
Answer: ________________

2. The circle below is cut into 3 equal pieces. One piece is shaded. What unit fraction of the circle is shaded?



Answer: ________________

3. This window pane is cut into 6 equal pieces. One piece is shaded. What unit fraction of the window is shaded?



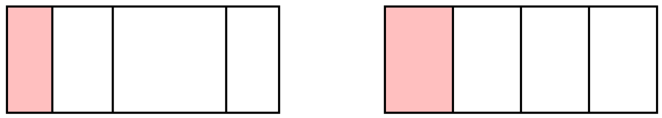
Answer: ___________________

4. A chocolate bar is cut into 8 equal pieces. One piece is shaded.

(a) What unit fraction of the bar is shaded? __________

(b) How many pieces that size are needed to fill the whole bar? __________

5. Both shapes below were cut into four pieces, and one piece of each is shaded. Only one of them shows *fourths*.

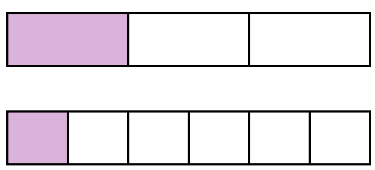


A B

(a) Circle the letter of the model that shows fourths: A B

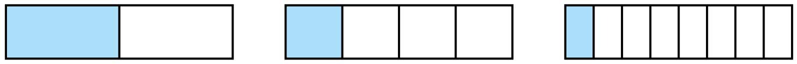
(b) Explain why the other model does not show fourths.

6. Two ribbons of the same length are cut into equal pieces: the first into 3 pieces and the second into 6 pieces. One piece of each is shaded.



Which shaded piece is bigger? Write >, <, or = in the blank: $\frac{1}{3}$ ________ $\frac{1}{6}$

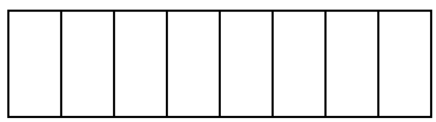
7. Three flags of the same size are cut into 2, 4, and 8 equal pieces. One piece of each flag is shaded.



Write the three shaded unit fractions in order, smallest first:

_________ _________ _________

8. A garden bed is divided into 8 equal plots, shown below.



(a) What unit fraction of the garden is one plot? _________

(b) Shade three plots. Then explain what fraction of the garden is shaded, and say how many copies of the unit fraction from part (a) that is.